Lecture 18

Holonomy and Gauss–Bonnet

Objectives. *Parallel transport around a loop can rotate a tangent vector. On a surface, the rotation is the enclosed curvature, modulo a full turn. Keeping track of the boundary tangent and its corner turns gives the Gauss–Bonnet formula. Summing that formula over a closed surface relates total curvature to its Euler characteristic. The ideal Foucault pendulum relates latitude holonomy to a slowly rotating swing line.*

By the end of this lecture you will be able to:

- Compute surface holonomy using a frame defined on the enclosed disk.
- Derive the angle excess and the boundary terms in local Gauss–Bonnet.
- Prove global Gauss–Bonnet given the cited topological inputs.
- Compute the latitude dependence of ideal Foucault precession.

18.1 Holonomy on an oriented surface

Let (M, g) be an oriented Riemannian surface with its Levi-Civita connection. Write $\mathrm{d}A$ for its metric area form. The notation $\mathrm{d}A$ names the area form, rather than the exterior derivative of a globally defined one-form A. In a positive orthonormal frame $e = (e_1, e_2)$, recall Lecture 17's convention

$$\nabla e_1 = -\omega^1{}_2 e_2, \qquad \nabla e_2 = \omega^1{}_2 e_1, \qquad \mathrm{d}\omega^1{}_2 = K\mathrm{d}A.$$

Definition 18.1. The positive quarter-turn J is defined by $Je_1 = e_2$ and $Je_2 = -e_1$.

It is independent of the positive orthonormal frame because plane rotations commute with this quarter-turn.

Definition 18.2. For a piecewise smooth loop γ based at p, its *holonomy* is the parallel-transport map of Lecture 16,

$$P_\gamma : T_pM \longrightarrow T_pM,$$

composed along its smooth segments.

Parallel transport preserves the metric and orientation, so holonomy is a rotation. To see the angle and its sign, write a transported unit vector along a framed segment as

$$V = \cos\alpha\, e_1 + \sin\alpha\, e_2, \qquad \frac{DV}{dt} = (\dot\alpha - \omega^1{}_2(\dot\gamma))JV.$$

The derivative follows by the product rule and the two frame actions above. The angle α has a real lift along the interval by Lecture 3's circle lifting theorem. Parallel transport means $\dot\alpha = \omega^1{}_2(\dot\gamma)$. Thus its matrix between the positive frames at the segment endpoints is the

rotation through $\alpha(b) - \alpha(a)$, measured counter-clockwise. The same construction works across finitely many curve segments. The figure illustrates the angle measured relative to a local frame.

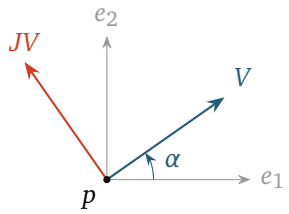


Figure 18.1: A unit vector has angle α from the first frame vector. Parallel transport changes this angle by the connection integral along the curve.

Theorem 18.3 (Surface holonomy · *proved given partitions of unity*)**.** *Let $D \subset M$ be a compact disk with smooth boundary, contained in an open set with one positive orthonormal frame. Parallel transport around ∂D, with its positive boundary orientation, rotates vectors by*

$$\Theta_D \equiv \int_D K \mathrm{d}A \pmod{2\pi}.$$

Proof. The frame agrees with itself when the loop returns to its starting point. Integrating the parallel-angle equation and using Stokes gives

$$\alpha(b) - \alpha(a) = \oint_{\partial D} \omega^1{}_2 = \int_D \mathrm{d}\omega^1{}_2 = \int_D K \mathrm{d}A.$$

A tangent-plane rotation records this real angle only modulo 2π. Linearity of transport extends the unit-vector conclusion to every vector. ∎

Reversing the loop negates the angle. A different positive frame along the loop changes the net lifted angle $\alpha(b) - \alpha(a)$ by an integer multiple of 2π. Indeed, its change-of-frame map is circle-valued, and its lifted angle has endpoint difference $2\pi m$ for some $m \in \mathbb{Z}$ by Theorem 3.11. The rotation is independent of that choice. Stokes requires a frame on the whole disk, not merely near its boundary.

For disks shrinking to p, choose the real holonomy angle tending to zero. The theorem identifies this lift with the curvature integral. Continuity gives

$$\left| \frac{\Theta_D}{\mathrm{Area}(D)} - K(p) \right| \leq \sup_{q \in D} |K(q) - K(p)| \longrightarrow 0.$$

Thus Gaussian curvature is the limiting rotation per unit area. Lecture 17's surface curvature operator is $R(e_1, e_2) = -KJ$. Its negative generates the infinitesimal holonomy rotation. For a positively traversed coordinate square of side ϵ, in coordinates orthonormal at p, the rotation expansion gives

$$P_{\square_\epsilon} = \mathbb{1} + \epsilon^2 K(p) J + O(\epsilon^3) = \mathbb{1} - \epsilon^2 \Omega_{\mathrm{curv}}(\partial_x, \partial_y)_p + O(\epsilon^3).$$

Here Ω_{curv} is Lecture 17's matrix of curvature two-forms, written Ω there. The subscript separates it from the solid angle Ω_D of §18.6. Lecture 21 proves the corresponding small-square formula for a general connection.

18.2 Geodesic triangles and angle excess

Definition 18.4. A *geodesic triangle* bounds an embedded disk by three regular geodesic segments, with nondegenerate corners. Its interior angles α_1, α_2 and α_3 lie in $(0, 2\pi)$.

These interior angles are distinct from the transported-vector angle α.

Definition 18.5. The signed *exterior turn* at a vertex is $\varepsilon_i = \pi - \alpha_i \in (-\pi, \pi)$. It is the change from the incoming to the outgoing tangent, positive counter-clockwise in a positive tangent frame.

Definition 18.6. The *angle excess* is $\alpha_1 + \alpha_2 + \alpha_3 - \pi$.

We use the cited *turning-angle theorem*: for the positive boundary of a disk in one positive frame extending over the disk, the continuous changes of tangent angle along the sides plus the exterior turns add to 2π. See Tu, Theorem 17.4, pp. 142–143. For the unit circle in a Cartesian frame, the positive unit tangent is $(-\sin t, \cos t)$, whose angle increases by 2π. This verifies the circle case. The general theorem, including nonconvex boundaries, remains a cited input. A frame defined only on an annulus does not satisfy its extension hypothesis. The latitude calculation in Example 18.16 compares zero tangent turning in the spherical frame with one full turn in a frame defined over the cap. For a proof in a chart, see Lee, Theorem 9.1 and Lemma 9.2, pp. 158–163.

Theorem 18.7 (Triangle excess · *proved given turning and partitions of unity*)**.** *If a geodesic triangle D lies in one framed neighbourhood as above, then*

$$\alpha_1 + \alpha_2 + \alpha_3 - \pi = \int_D K \mathrm{d}A.$$

Proof. Along each side, the unit tangent is parallel, so its frame angle changes by the integral of $\omega^1{}_2$ on that side. The turning theorem therefore gives

$$\oint_{\partial D} \omega^1{}_2 + \sum_i (\pi - \alpha_i) = 2\pi.$$

Stokes gives $\oint_{\partial D} \omega^1{}_2 = \int_D K \mathrm{d}A$. Rearranging proves the formula. For this piecewise smooth boundary, Stokes follows by rounding each corner in a coordinate ball of radius ϵ_{r}. Smoothness bounds the changed connection integral by $O(\epsilon_{\mathrm{r}})$ and the changed curvature integral by $O(\epsilon_{\mathrm{r}}^2)$. The replacement arcs have length $O(\epsilon_{\mathrm{r}})$, including at concave corners. Apply smooth-boundary Stokes to the rounded disk and take $\epsilon_{\mathrm{r}} \to 0$. ∎

Example 18.8 (Angle excess of the octant)**.** Lecture 17 transported $e_x \in T_N S^2$ around the octant boundary $N \to e_x \to e_y \to N$ in Fig. 17.1. It returned as e_y, a rotation by $+\pi/2$ in the outward orientation. The unit-sphere octant has curvature integral

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{\pi}{2}.$$

Its three right angles give the same excess $3\pi/2 - \pi = \pi/2$. This triangle is contained in a framed chart obtained by removing a point outside the octant from the sphere.

18.3 The local Gauss–Bonnet formula

Let $T = \dot\gamma$ be the unit tangent of a smooth boundary segment parametrized by arc length s.

Definition 18.9. The signed *geodesic curvature* of this segment is

$$k_g = g(\nabla_T T, JT).$$

For a positive boundary, JT points into the disk. Differentiating $g(T,T) = 1$ makes its covariant acceleration perpendicular to T. Writing $T = \cos\zeta\, e_1 + \sin\zeta\, e_2$ in the angle derivative already computed gives

$$\nabla_T T = (\zeta' - \omega^1{}_2(T))JT, \qquad k_g = \zeta' - \omega^1{}_2(T).$$

Thus geodesic sides have zero geodesic curvature. The corner diagram distinguishes a jump in the tangent from curvature on a smooth side.

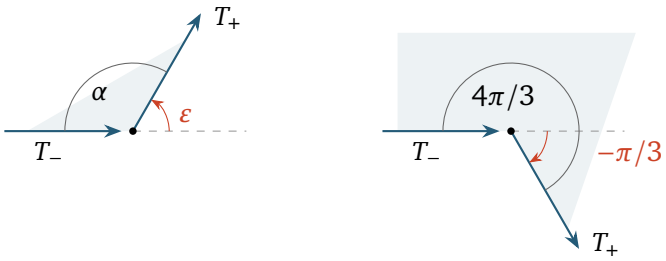


Figure 18.2: The exterior turn ε takes the incoming tangent to the outgoing tangent. The interior angle is $\alpha = \pi - \varepsilon$. The concave panel has $\alpha = 4\pi/3$ and exterior turn $-\pi/3$.

Theorem 18.10 (Local Gauss–Bonnet · *proved given turning and partitions of unity*)**.** *Let D be a compact embedded disk whose positive boundary has finitely many regular smooth sides and nondegenerate corners. Suppose one positive orthonormal frame is defined on a neighbourhood of all of D. Then*

$$\int_D K\,\mathrm{d}A + \oint_{\partial D} k_g\,\mathrm{d}s + \sum_i \varepsilon_i = 2\pi.$$

The side integrals exclude the corners. Their turns occur in the finite sum.

Proof. Integrate $k_g = \zeta' - \omega^1{}_2(T)$ on each side. The turning theorem gives $\sum \Delta\zeta + \sum_i \varepsilon_i = 2\pi$. Summing the side integrals therefore gives

$$\oint_{\partial D} k_g\,\mathrm{d}s = 2\pi - \sum_i \varepsilon_i - \oint_{\partial D} \omega^1{}_2.$$

Use Stokes, with the same corner-rounding limit as above, to replace the last integral by $\int_D K\,\mathrm{d}A$. ∎

Consider a northern unit-sphere cap $0 \le \theta \le a$, where $0 < a < \pi$. Its positive boundary runs in increasing ϕ. Use the unit spherical frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The unit tangent is $T = e_\phi$ and $JT = -e_\theta$. Lecture 16's symbols give $\nabla_T T = -\cot a\, e_\theta$, so $k_g = \cot a$. The two integrals are

$$\int_D K\,\mathrm{d}A = 2\pi(1 - \cos a), \qquad \oint_{\partial D} k_g\,\mathrm{d}s = \cot a\,(2\pi \sin a) = 2\pi\cos a.$$

Their sum is 2π, with no corner contribution. A smooth frame on this cap is exhibited in the latitude example below.

18.4 Gauss–Bonnet on a closed surface

Theorem 18.11 (Gauss–Bonnet · *proved given cited topology and Stokes inputs*)**.** *Every closed oriented Riemannian surface satisfies*

$$\int_M K \mathrm{d}A = 2\pi\chi(M).$$

Proof. We apply local Gauss–Bonnet to each triangular face and add. Shared-edge integrals cancel, and the angles at each vertex fill one full turn. The remaining count is $V - E + F$. We use a finite chart-contained smooth triangulation, with a compatible Δ-complex structure as in Lecture 6. The construction after this proof justifies these inputs and the hypotheses of the local formula.

Let V, E and F count vertices, edges and faces. Orient each face by the surface orientation and sum the local formula. Every shared edge has opposite boundary directions in its two faces. Reversing T leaves $\nabla_T T$ unchanged and negates JT, so it negates k_g. The two edge integrals cancel. At each vertex, the incident interior angles sum to 2π. Thus the sum of exterior turns over all face corners is $3\pi F - 2\pi V$. The figure displays these cancellations and the complete angle around a vertex.

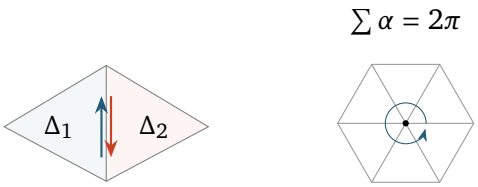


Figure 18.3: Adjacent faces traverse a shared edge in opposite directions. The incident face angles at an interior vertex fill one full turn.

The summed formula gives

$$\int_M K \mathrm{d}A = 2\pi F - (3\pi F - 2\pi V) = 2\pi V - \pi F = 2\pi(V - E + F).$$

The last equality uses $3F = 2E$, since each triangular side belongs to two faces on a surface without boundary. By Theorems 6.13 and 6.14, $V - E + F = \chi(M)$. For disconnected M, apply the argument to each component and add. ∎

Triangulations and the local hypotheses. Use the cited existence of a finite smooth triangulation of a compact surface (Lee, discussion before Theorem 9.7, pp. 166–167). In Lee's definition, each curved triangle and its enclosed region lie in one coordinate chart, with regular sides and no cusps (p. 162). We also use the cited planar *Schoenflies theorem*: a simple closed plane curve and its enclosed region form a closed topological disk (Stillwell, §0.3.9, pp. 35–36). Gram–Schmidt on the connected chart neighbourhood of the face gives a positive orthonormal frame, changing one vector's sign if needed. Thus each closed face is a disk to which the local formula applies. The edges need not be geodesics.

To obtain the required Δ-complex, order the vertices and choose one parametrization for each edge, from its lower to its higher vertex. These give a homeomorphism from the boundary of each standard triangle to its face boundary. Identify the standard triangle with a closed unit disk by radially rescaling rays from its centroid. Schoenflies supplies a disk homeomorphism. It can be chosen to match that boundary map. Indeed, any difference

on the boundary circle is a homeomorphism $\psi\colon S^1 \to S^1$, which extends to the disk by $r\zeta \mapsto r\psi(\zeta)$, with the centre fixed. Composing with this extension matches the chosen boundary map. The face maps now agree on shared edges and give a finite Δ-complex structure on M.

The integral does not change when the metric changes, although the pointwise Gaussian curvature and the area form both change. A sphere therefore has total curvature 4π for every Riemannian metric. A torus has total curvature zero for every Riemannian metric, even when its curvature is not identically zero. For the closed oriented genus-g surface Σ_g, Lecture 6 gives $\chi(\Sigma_g) = 2 - 2g$. Hence

$$\int_{\Sigma_g} K\,\mathrm{d}A = 4\pi(1-g).$$

The figure compares an embedded torus with a surface homeomorphic to the sphere. The colours distinguish regions of positive and negative Gaussian curvature.

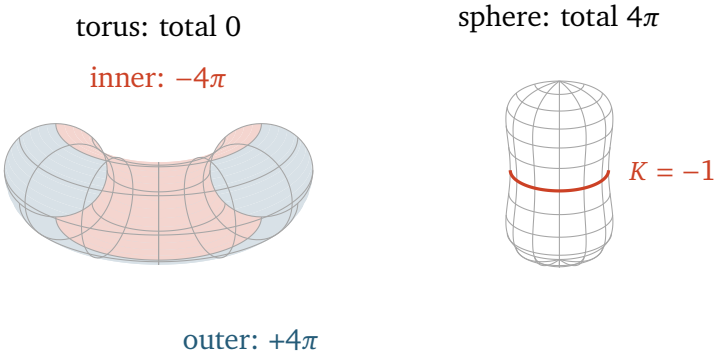


Figure 18.4: For the cutaway torus with $b = 2$ and $r = 1$, the outer and inner halves contribute $+4\pi$ and -4π. The surface with a waist has total curvature 4π and curvature -1 at its equator.

Example 18.12 (Curvature integrals on the embedded torus)**.** We use the torus of Example 17.24, with $b > r > 0$ and its chart orientation (v, u). Its positive area form is $\mathrm{d}A = r(b + r\cos v)\mathrm{d}v \wedge \mathrm{d}u$. That example found the Gaussian curvature. Multiplying by the area form gives

$$K = \frac{\cos v}{r(b + r\cos v)}, \qquad K\mathrm{d}A = \cos v\,\mathrm{d}v \wedge \mathrm{d}u.$$

Integrating over $-\pi/2 < v < \pi/2$ and one full u-circle gives $+4\pi$. Integrating over $\pi/2 < v < 3\pi/2$ and one full u-circle gives -4π. The two contributions cancel, although neither vanishes separately.

For the sphere panel, we revolve the meridian $(f(t), z(t))$, given by

$$f(t) = \sin t\big(1 - \frac{1}{2}\sin^2 t\big), \qquad z(t) = \cos t, \qquad 0 \le t \le \pi.$$

The surface is smooth at the two poles and has no self-intersections. To compute its curvature, put $h = \sqrt{f'^2 + z'^2}$. On $0 < t < \pi$, the coframe $(h\mathrm{d}t, f\mathrm{d}u)$ gives $\omega^1{}_2 = -(f'/h)\mathrm{d}u$. Thus

$$K = -\frac{1}{hf}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{f'}{h}\right).$$

At the waist $t = \pi/2$, the values are $f = 1/2$, $f' = 0$, $f'' = 1/2$ and $h = 1$, so $K = -1$. Gauss–Bonnet still gives total curvature 4π because the surface is homeomorphic to S^2.

18.5 The Euler number of the tangent sphere

Recall from §14.5 that the index of an isolated tangent-field zero is the winding of its normalized components on a small positive circle. The cited Poincaré–Hopf theorem, Theorem 14.9, states that the sum of these indices on a closed oriented surface is its Euler characteristic. In particular this sum is independent of the chosen field with isolated zeros.

Definition 18.13. For the tangent bundle of S^2, call this common sum its *Euler number*.

Corollary 18.14 (The tangent-sphere Euler number · *sketched given Poincaré–Hopf*)**.** *The Euler number of TS^2 is $\chi(S^2) = 2$.*

The field $X(p) = e_3 - p_3 p$ from Lecture 14 has its two zeros at the poles, with index +1 at each. It supplies a field to which the definition applies. Poincaré–Hopf makes every such index sum equal to 2. A nowhere-zero tangent field would have empty index sum zero, contradicting this value. Hence there can be no global tangent frame on the sphere. A disk in one coordinate chart has a framed neighbourhood by Gram–Schmidt. The local curvature formula requires that the chosen frame extend over that disk, rather than only around its boundary. The whole sphere admits no such frame. We can also compare the index sum directly with the curvature integral. Let X be a smooth tangent field on a closed oriented surface, with all zeros isolated. Its zero set is closed and discrete, hence finite by compactness. Choose disjoint small disks D_i around its zeros p_i, with positive orthonormal frames f_i defined over the disks. On $M' = M \setminus \bigcup_i \operatorname{int}(D_i)$ the vectors $e_1 = X/|X|$ and $e_2 = Je_1$ give a positive orthonormal frame. Write ω_X for its connection entry. Each boundary component of M' has the opposite orientation to ∂D_i. Stokes therefore gives

$$\int_{M'} K \mathrm{d}A = -\sum_i \oint_{\partial D_i} \omega_X .$$

Near ∂D_i, let ψ_i be a local lift of the angle from $f_{i,1}$ to e_1. Differentiating this frame rotation gives $\omega_X = \omega_{f_i} - \mathrm{d}\psi_i$. The angle differential agrees on overlapping lifts. Its integral is $2\pi \operatorname{ind}_{p_i} X$ by the definition of index. Stokes in the disk frame now gives

$$\oint_{\partial D_i} \omega_X = \int_{D_i} K \mathrm{d}A - 2\pi \operatorname{ind}_{p_i} X .$$

Adding the disk integrals to the punctured-surface integral yields

$$\frac{1}{2\pi} \int_M K \mathrm{d}A = \sum_i \operatorname{ind}_{p_i} X .$$

Thus the Euler number of TS^2 is also its total curvature divided by 2π. Together with Gauss–Bonnet, this argument proves the surface case of Lecture 14's cited Poincaré–Hopf theorem. Lecture 22 expresses the same integer using changes between local frames.

18.6 Latitude holonomy and the Foucault pendulum

Definition 18.15. The *solid angle* of a region on the unit sphere is its area.

Let the latitude be $\lambda \in (-\pi/2, \pi/2)$ and put $a = \pi/2 - \lambda$. Follow its circle once in increasing ϕ.

Example 18.16 (Latitude rotation in a cap frame)**.** Recall the transport around this latitude from Example 16.22. Its angle change in the spherical frame was $-2\pi\cos a = -2\pi\sin\lambda$. The northern cap's curvature integral is its solid angle,

$$\Omega_D = 2\pi(1-\cos a) = 2\pi(1-\sin\lambda).$$

This differs from Lecture 16's angle by 2π and represents the same holonomy. The spherical frame itself is undefined at the north pole, so Stokes cannot be applied to that frame across the cap.

To use a frame on the full cap, define the rotated vectors by

$$e_{N,1} = \cos\phi\, e_\theta - \sin\phi\, e_\phi, \qquad e_{N,2} = \sin\phi\, e_\theta + \cos\phi\, e_\phi.$$

We substitute $e_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta)$ and $e_\phi = (-\sin\phi, \cos\phi, 0)$. The identity

$$\frac{1-\cos\theta}{\sin^2\theta} = \frac{1}{1+\cos\theta}$$

removes the apparent singularity at the north pole. In Cartesian coordinates $p = (x, y, z)$ on the unit sphere, the columns are

$$E_N = \begin{pmatrix} 1 - x^2/(1+z) & -xy/(1+z) \\ -xy/(1+z) & 1 - y^2/(1+z) \\ -x & -y \end{pmatrix}.$$

The rotation makes it tangent, positive and orthonormal away from the poles. The displayed expression is smooth for $z > -1$ and at the north pole gives e_x and e_y. Continuity therefore preserves those properties there. Differentiating the rotated frame and using Lecture 17's spherical-frame form $\omega^1{}_2 = -\cos\theta\mathrm{d}\phi$ gives

$$\omega_N = \omega^1{}_2 + \mathrm{d}\phi = (1-\cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}.$$

This is smooth at the north pole. Its integral around the latitude is $2\pi(1-\cos a)$, so Stokes in this frame gives the cap's curvature integral. At the equator the holonomy is zero modulo 2π, while the cap integral is 2π. Holonomy alone does not retain that full turn.

On a positive latitude boundary, the unit tangent is $T = e_\phi$. Its components in the spherical frame are constant, so its tangent angle has total change zero. This frame does not extend to the north pole. In the smooth north frame, the same tangent is

$$T = -\sin\phi\, e_{N,1} + \cos\phi\, e_{N,2}.$$

Its lifted angle is $\phi + \pi/2$, with total change $+2\pi$. This verifies the extension hypothesis in the turning-angle theorem for this cap. The dials in the figure compare the tangent components in these two frames.

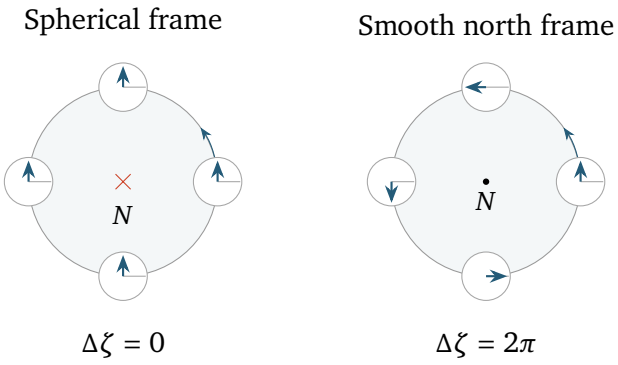


Figure 18.5: The same positively traversed latitude has constant tangent components in the singular spherical frame and one positive turn in the smooth north frame.

Physical input. We use the ideal small-swing pendulum model on a slowly rotating sphere. For the sphere's angular velocity vector $\boldsymbol{\nu}_\oplus$, the rotating-frame Coriolis acceleration is $-2\boldsymbol{\nu}_\oplus \times v$. For a small horizontal displacement ξ, the restoring acceleration is $-\nu_0^2\xi$, where $\nu_0 > 0$ is the natural angular frequency. The model retains terms linear in the horizontal displacement and the rotation rate. It neglects damping, anisotropy, centrifugal corrections and vertical motion at this order.[1]

Let $\nu_\oplus > 0$ be the sphere's angular rotation rate, with $\nu_\oplus \ll \nu_0$. In a positive east–north frame, write the small displacement as $\xi = \xi_E + i\xi_N$. Let n be the outward unit normal. The angular velocity decomposes as

$$\boldsymbol{\nu}_\oplus = (\boldsymbol{\nu}_\oplus \cdot n)n + \boldsymbol{\nu}_{\oplus,\tan}, \qquad b = \boldsymbol{\nu}_\oplus \cdot n = \nu_\oplus \sin\lambda.$$

For horizontal v, the cross product of $\boldsymbol{\nu}_{\oplus,\tan}$ with v is normal. Also $n \times v = Jv$ in the positive east–north frame. Hence the horizontal projection is

$$(-2\boldsymbol{\nu}_\oplus \times v)_{\tan} = -2bJv.$$

The figure shows the normal component and the resulting acceleration.

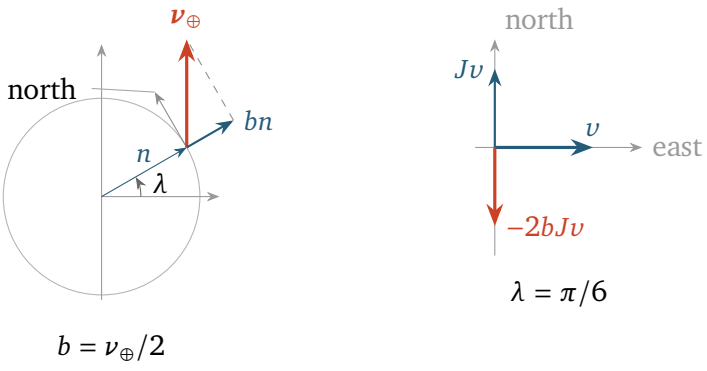


Figure 18.6: At latitude $\lambda = \pi/6$, the axis has normal component $\nu_\oplus/2$. In the positive east–north plane, an eastward velocity has southward Coriolis acceleration.

[1] See Fitzpatrick, *Newtonian Dynamics*, Foucault Pendulum, for the linear Coriolis model, and Dray, *The Foucault Pendulum* (1998), for its slowly changing direction and tangential-projection interpretation.

Multiplication by i represents J on $\xi_E + i\xi_N$. Adding restoring and Coriolis accelerations gives

$$\ddot{\xi} + 2ib\dot{\xi} + \nu_0^2 \xi = 0.$$

Substituting $\xi = e^{-ibt} w$ cancels the first-derivative terms and gives

$$\ddot{w} + (\nu_0^2 + b^2) w = 0.$$

The b^2 term is an exact consequence of solving this linear model. It does not account for all physical corrections of second order in rotation. An oscillation along a fixed line in the w plane therefore has slow orientation angle $-bt$. Its rate agrees with parallel transport:

$$\dot{\alpha} = -\cos a \, \dot{\phi} = -\nu_\oplus \sin \lambda = -b.$$

Over one rotation $2\pi/\nu_\oplus$, the change is $-2\pi \sin \lambda$. It is clockwise in the northern hemisphere and counter-clockwise in the southern. At latitude $\lambda = \pi/6$, the model predicts $-\pi$ in one rotation of the sphere. The oriented direction reverses while the swing line returns.

A geometric heuristic interprets the slowly varying swing direction after averaging over its fast oscillations. In this idealization, no force turns the swing line within the moving horizontal plane. Its ambient change is only the normal change needed to stay tangent to the sphere. By Lecture 16's tangential-projection description, this is parallel transport. The Coriolis calculation above supplies the rate within the stated model. An oriented swing direction records angles modulo 2π. The physical unoriented swing line records them modulo π. For a spin-1/2 ground state aligned with a field direction traversing this latitude, Lecture 24 derives the Berry phase $-\Omega_D/2$ modulo 2π.

Self-check. Check global Gauss–Bonnet for the round sphere and the flat quotient torus. Decide whether zero total curvature forces pointwise zero curvature.

Answer. On a sphere of radius R, curvature $1/R^2$ times area $4\pi R^2$ gives $4\pi = 2\pi\chi(S^2)$. On the flat torus $K = 0$, so the integral is $0 = 2\pi\chi(T^2)$, using $\chi(T^2) = 0$ from Lecture 6. Zero total curvature does not force pointwise zero curvature. The embedded torus above has positive outer curvature and negative inner curvature, whose integrals are $+4\pi$ and -4π.

Self-check. On the unit sphere, take the sector $0 \le \theta \le \arccos(2/5)$ and $0 \le \phi \le 3\pi/5$. Compute its curvature integral, its smooth-side geodesic-curvature integral and its total exterior turn. Verify local Gauss–Bonnet and give its holonomy angle.

Answer. Put $a = \arccos(2/5)$. The area integral is $(3\pi/5)(1 - 2/5) = 9\pi/25$. The meridians have zero geodesic curvature. On the latitude arc, the integral is

$$\int_0^{3\pi/5} \cot a \sin a \, d\phi = \frac{3\pi}{5}\frac{2}{5} = \frac{6\pi}{25}.$$

The interior angles are $3\pi/5$ at N and $\pi/2$ at each latitude endpoint. Thus the exterior turns sum to $2\pi/5 + \pi/2 + \pi/2 = 7\pi/5$. Their sum with the two integrals is 2π. The holonomy is $9\pi/25$ modulo 2π by the curvature integral.

Self-check. State in one sentence what holonomy is and why we introduced it. Explain why a holonomy rotation alone cannot recover the full curvature integral.

Answer. Holonomy is parallel transport around a based loop as a map $P_\gamma : T_pM \to T_pM$, introduced to compare a vector with its return and measure curvature through the resulting

rotation. A rotation retains only its angle modulo 2π. For example, an equator has identity holonomy although its northern cap has curvature integral 2π.

Summary. Surface transport has angle $\int_D K\mathrm{d}A$ modulo 2π when one positive orthonormal frame covers the disk. The turning theorem adds the boundary tangent and corner turns, giving local Gauss–Bonnet. Using the cited topology inputs and summing over the triangulation yields $\int_M K\mathrm{d}A = 2\pi\chi(M)$ on every closed oriented Riemannian surface. The tangent sphere has Euler number 2 by the cited Poincaré–Hopf theorem. Latitude transport gives the ideal Foucault angle $-2\pi\sin\lambda$ modulo 2π. Lecture 19 uses curvature and geodesics in a Lorentzian spacetime.

Reading. Tu, *Differential Geometry*, §17, especially Theorems 17.4–17.5 and 17.9, pp. 142–146. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Ch. 9, pp. 158–168, including the smooth triangulation discussion. Stillwell, *Classical Topology and Combinatorial Group Theory*, §0.3.9, pp. 35–36 (Schoenflies theorem).